

William Fowler

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EDUCATION

Tufts University, Medford, MA September 2022 – May 2026

B.S. in Computer Science, Summa Cum Laude — GPA: 3.95

Relevant Courses: Trusted and Responsible AI, Artificial Agents & Autonomy, Reinforcement Learning, Intro to Machine Learning

BlueDot Impact, virtual June 2024 – September 2024

AI Safety Fundamentals Course

EXPERIENCE

ERA Fellowship, Cambridge, UK (remote) June 2026 – Present

Research Manager (part-time)

- Manage residents in the inaugural Frontier AI Security Residency, scoping hardware verification research projects
- Review ~1,500 applicants across cyber, hardware, and fieldbuilding tracks

Existential Risk Laboratory (XLab), Chicago, IL June 2026 – Present

Research Fellow

- Develop verification techniques estimating LLM training FLOPs from power draw, memory bandwidth, and inter-GPU traffic
- Design adversarial training runs to red-team FLOP verification techniques

Cambridge Boston Alignment Initiative, Boston, MA (remote) November 2025 – June 2026

Research Assistant

- Ran a validation study benchmarking LLM classification of AI incidents against human expert consensus across five taxonomies; final model and prompt configurations matched or exceeded human performance
- Handed off the improved pipeline to a full-time researcher; results published on the MIT AI Risk Repository blog

Pivotal Research, London, UK February 2025 – April 2025

Technical AI Governance Fellow

- Researched the technical feasibility and progress of decentralized training for frontier AI models
- Authored a report translating decentralized training constraints (bandwidth, hardware) for a policy audience

The University of Chicago, Chicago, IL Jun 2023 – Dec 2023, Jun 2024 – Aug 2024

Research Assistant

- Conducted research comparing autonomous driving algorithms, yielding two peer-reviewed publications (see below)
- Designed and implemented a data pipeline for automatic weather stations; built educational materials for ML coursework

PROJECTS

Estimating FLOPs on the Jetson Orin Nano March 2026 – July 2026

- Reduced estimator error below 10% across all 21 tested LLM training workloads
- Built a methodology for estimating training FLOPs using only system-level metrics, without access to model code or logs
- Write-up: [lesswrong.com: Estimating LLM Training FLOPs on the NVIDIA Jetson Orin Nano](https://lesswrong.com/Estimating-LLM-Training-FLOPs-on-the-NVIDIA-Jetson-Orin-Nano)

Improving the AI Incident Tracker (MIT FutureTech) November 2025 – June 2026

- Refined prompts and model selection to meet or outperform expert human classification across all 5 covered taxonomies
- Write-up: [airisk.mit.edu: AI Incident Tracker June 2026 Update](https://airisk.mit.edu/AI-Incident-Tracker-June-2026-Update)

PUBLICATIONS

W. Fowler, A. Esquivel Morel, K. Keahey. "Encoding Consistency: Optimizing Self-Driving Reliability With Real-Time Speed Data." In Proceedings of the 4th Workshop on Flexible Resource and Application Management on the Edge (FRAME '24), pp. 47–50. 2024.

A. Esquivel Morel, W. Fowler, K. Keahey, K. Zheng, M. Sherman, R. Anderson. "AutoLearn: Learning in the Edge to Cloud Continuum." In Proceedings of the SC '23 Workshops (SC-W '23), pp. 350–356. ACM. <https://doi.org/10.1145/3624062.3624101>

TECHNICAL SKILLS

Python, C/C++, PyTorch, CUDA, Transformers, Linux/embedded (NVIDIA Jetson)

ACTIVITIES

AI Safety Student Team (AISST), Technical Member – reviewed AI policy and safety literature Sept 2024 – May 2026

Chuckleducks Improv Comedy, Treasurer – managed the ~\$2,000 annual budget Nov 2022 – May 2026

Tufts National Student Data Corps, Co-President – led comms and mentorship for 87 members Sept 2022 – Jan 2025

ADDITIONAL EXPERIENCE

Tufts University, Medford, MA

September 2024 – May 2026

Teaching Assistant

- Instructed a lab of ~20 students, guiding hands-on exercises in C and assembly language programming
- Held biweekly office hours providing one-on-one assistance with homework projects in C
- Evaluated and graded student project design documents, offering constructive feedback

Subject Tutor (January 2023 – December 2024)

- Tutored Tufts students in Intro to Computer Science, Data Structures, Calculus 2, and Calculus 3
- Moderated small group meetings for students seeking help in their classes

Volunteer Tutor

November 2022 – May 2023

- Mentored one seventh-grade student in weekly sessions
- Created study plans for Math, Language Arts, Geography, and Science

Code Ninjas Alexandria, Alexandria, VA

June 2021 – August 2022

Code Sensei

- Taught group and one-on-one coding lessons to kids aged 6–14
- Graded students' JavaScript and C# projects
- Directed weekly summer camp activities for around 12 campers